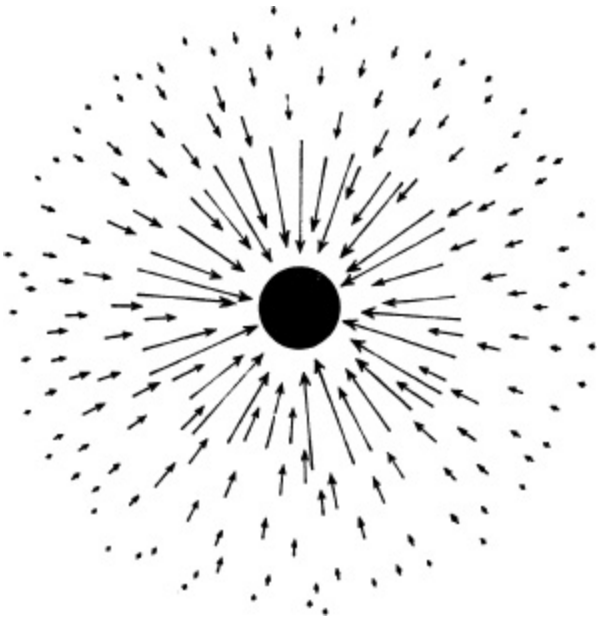
now decelerating into the vicinity of the black hole closest to Earth, a hole called “Hades” near the star Vega.



P.1 Atoms of gas, pulled by a black hole’s gravity, stream toward the hole from all directions.

On your ship’s video screen you and your crew see evidence of the hole’s presence: The atoms of gas that sparsely populate interstellar space, approximately one in each cubic centimeter, are being pulled by the hole’s gravity (Figure P.1). They stream toward the hole from all directions, slowly at great distances where gravity pulls them weakly, faster nearer the hole where gravity is stronger, and extremely fast—almost as fast as light—close to the hole where gravity is strongest. If something isn’t done, your starship too will be sucked in.

Quickly and carefully your first mate, Kares, maneuvers the ship out of its plunge and into a circular orbit, then shuts off the engines. As you coast around and around the hole, the centrifugal force of your circular motion holds your ship up against the hole’s gravitational pull. Your ship is like a toy slingshot of your youth on the end of a whirling string, pushed out by its centrifugal force and held in by the string’s tension, which. is like the hole’s gravity. As the starship coasts, you and your crew prepare to explore the hole.